

DEPRESSION

SCORING MANUAL

A brief guide to scoring the PROMIS® Depression instruments:

ADULT	PEDIATRIC	EARLY CHILDHOOD PARENT-REPORT	PARENT PROXY
PROMIS Item Bank v1.0 – Depression PROMIS Item Bank v1.0 – Depression (recommended) PROMIS Item Bank v1.0 – Depression (screen-to-CAT) PROMIS Short Form v1.0 – Depression 4a PROMIS Short Form v1.0 – Depression 6a PROMIS Short Form v1.0 – Depression 8a PROMIS Short Form v1.0 – Depression 8b PROMIS-Ca Bank v1.0 – Depression PROMIS Short Form v1.0 – Depression-OA-Knee 4a	PROMIS Pediatric Bank GenPop v3.0 – Depressive Symptoms PROMIS Pediatric Bank GenPop v3.0 – Depressive Symptoms (screen-to-CAT) PROMIS Pediatric Short Form GenPop v3.0 – Depressive Symptoms 8a PROMIS Pediatric Item Bank v2.0 – Depressive Symptoms* PROMIS Pediatric Item Bank v2.0 – Depressive Symptoms (recommended)* PROMIS Pediatric Item Bank v2.0 – Depressive Symptoms (screen-to-CAT)* PROMIS Pediatric Short Form v2.0 – Depressive Symptoms 8a* PROMIS Pediatric Item Bank v1.1 – Depressive Symptoms* PROMIS Pediatric Short Form v1.1 – Depressive Symptoms 8b* PROMIS Pediatric Item Bank v1.0 – Depressive Symptoms* PROMIS Pediatric Short Form v1.0 – Depressive Symptoms 8a*	PROMIS Early Childhood Parent-Report Bank v1.0 – Depressive Symptoms PROMIS Early Childhood Parent-Report Short Form v1.0 – Depressive Symptoms 4a PROMIS Early Childhood Parent-Report Short Form v1.0 – Depressive Symptoms 8a	PROMIS Parent Proxy Bank GenPop v3.0 – Depressive Symptoms PROMIS Parent Proxy Bank GenPop v3.0 – Depressive Symptoms (screen-to-CAT) PROMIS Parent Proxy Short Form GenPop v3.0 – Depressive Symptoms 6a PROMIS Parent Proxy Item Bank v2.0 – Depressive Symptoms* PROMIS Parent Proxy Item Bank v2.0 – Depressive Symptoms (recommended)* PROMIS Parent Proxy Item Bank v2.0 – Depressive Symptoms (screen-to-CAT)* PROMIS Parent Proxy Short Form v2.0 – Depressive Symptoms 6a* PROMIS Parent Proxy Item Bank v1.1 – Depressive Symptoms* PROMIS Parent Proxy Short Form v1.1 – Depressive Symptoms 6b* PROMIS Parent Proxy Item Bank v1.0 – Depressive Symptoms* PROMIS Parent Proxy Short Form v1.0 – Depressive Symptoms 6a*

*Retired measure

Ca=Cancer, GenPop=general population, OA=osteoarthritis

COMPARING SCORES ACROSS VERSIONS

Some PROMIS domains have multiple versions of instruments (i.e., v1.0, v1.1, v2.0, v3.0). Generally, **it is recommended that you use the most recent version available which can be identified as the instrument with the highest version number.** In most cases, an instrument that has a decimal increase (v1.0 to v1.1) retains the same item-level parameters as well as instrument reliability and validity. In cases where a version number increases by a whole number (e.g., v1.0 to v2.0), the changes to the instrument are more substantial.

- For Depression, the calibrations between v1.0, v1.1, and v2.0 are identical. Consequently, T-scores from v1.0, v1.1, and v2.0 are comparable.
- Scores from Pediatric and Parent Proxy GenPop v3.0 are NOT comparable with v1.0, v1.1, or v2.0 measure scores.

Transforming Scores between Versions:

To transform T-scores from Pediatric/Parent Proxy v1.0/v1.1/v2.0 Depressive Symptoms measures to the Pediatric/Parent Proxy GenPop v3.0 metric, rescore v1.0/v1.1/v2.0 measures using the [HealthMeasures Scoring Service](#) or Assessment Center API. This is the preferred strategy as it produces precise scores.

1. Download the v3.0 measure from [Search & View Measures](#) on HealthMeasures.net.
2. Revise your dataset as needed to match the v3.0 item IDs and response scores:
 - **PROMIS Parent Proxy Bank GenPop v3.0 - Depressive Symptoms and PROMIS Parent Proxy Bank GenPop v3.0 - Depressive Symptoms (screen-to-CAT):**
 - The Item ID **Pf2depr1r** was changed to **Pf2depr1r2** to account for the revised Response Scores from 1-2-3-4-5 to 1-2-3-4-4.
3. Follow the instructions for using the [HealthMeasures Scoring Service](#). Contact help@HealthMeasures.net if using the Assessment Center API for rescoring.

The crosswalk tables in Appendix 3 may also be used as a reference to guide understanding of how scores have shifted from the older v1.0, v1.1, and v2.0 measures to the v3.0 measures. If these tables are used to convert older T-scores to GenPop v3.0, please note that the resulting T-scores are not as precise as those generated by using HealthMeasures Scoring Service or Assessment Center API.

WHICH CALIBRATION SAMPLE SHOULD I USE?

Some PROMIS Parent Proxy v1.0, v1.1, and v2.0 measures (Anxiety, Depressive Symptoms, Fatigue, Mobility, Pain Interference, Peer Relationships) had two calibration samples – “Parent Proxy” and “Parent Proxy Without Local Dependence.” The former (Parent Proxy) included calibrations for all items. This was the default calibration sample. The Parent Proxy Without Local Dependence did not include calibrations for some items. The items without calibrations are enemy items. That is, a dyad or triad of items was identified in which there are psychometric reasons to only administer one of those items to a given respondent. For example, item Pf2anxiety5 and Pf2anxiety4 are enemy items. A participant should only see one of these items in a CAT. The v3.0 GenPop measure have a single calibration sample and no enemy items.

SCORING THE INSTRUMENT

Short Forms: PROMIS instruments are scored using item-level calibrations. This means that the most accurate way to score a PROMIS instrument is to use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoring-service) or a data collection tool that automatically calculates scores (e.g., REDCap auto-score). This method of scoring uses responses to each item for each participant. We refer to this as “response pattern scoring.” Because response pattern scoring is more accurate than the use of raw score/scale score look up tables included in this manual, it is preferred. Response pattern scoring is especially useful when there is missing data (i.e., a respondent skipped an item), different groups of participants responded to different items, or you have created a new questionnaire using a subset of questions from a PROMIS item bank.

Instructions for the HealthMeasures Scoring Service

1. Download the Input Template from the [HealthMeasures Scoring Service](#) home page.
2. Add your data (all respondents’ individual answers to questions) for one PROMIS measure to the template (multiple timepoints are okay).

3. Determine how much data is missing. In order to produce a precise score, a respondent must answer at least 4 items in measures that include 4 or more items. If a precise score is required, remove respondents who answered less than 4 items from the input template and do not calculate a T-score.
4. Upload your spreadsheet to the HealthMeasures Scoring Service. If this is your first time, register as a new user.
5. Access your email to receive a spreadsheet with calculated scores.

Instructions for Using Raw Sum Score to T-score Conversion Tables

Each question usually has five response options ranging in value from one to five. To find the total raw score for a short form with all questions answered, sum the values of the response to each question. For example, for the adult 8-item form, the lowest possible raw score is 8; the highest possible raw score is 40 (see all short form scoring tables in Appendix 1). **All questions must be answered in order to produce a valid score using the scoring tables.** If a participant has skipped a question, use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoringervice) to generate a final score.

Locate the applicable score conversion table in Appendix 1 and use this table to translate the total raw score into a T-score for each participant. The T-score rescales the raw score into a standardized score with a mean of 50 and a standard deviation (SD) of 10. Therefore, a person with a T-score of 40 is one SD below the mean.

For the Adult PROMIS Depression 8b short form, a raw score of 10 converts to a T-score of 46.2 with a standard error (SE) of 2.8 (see scoring table for the 8b short form in appendix). Thus, the 95% confidence interval around the observed score ranges from 40.7 to 51.7 (T-score $\pm$ (1.96*SE) or $46.2 \pm (1.96*2.8)$).

CAT: A minimum number of items (e.g., 4) must be answered in order to receive a score for the Depression CATs. The response to the first item will guide the system's choice of the next item for the participant. The participant's response to the second item will dictate the selection of the following question, and so on. As additional items are administered, the potential for error is reduced and confidence in the respondent's score increases. The CAT will continue until either the standard error drops below a specified level (e.g., on the T-score metric 3.0), or the participant has answered the maximum number of questions (e.g., 12), whichever occurs first. For some CATs, specifically "recommended" and "screen-to-CAT" there are additional stopping rules. These include stopping when the standard error isn't improving much or if a respondent is asymptomatic. For details on the exact stopping rules for Depression CATs, view the Measure Differences summary.

SCORES

For most PROMIS instruments, a score of 50 is the average for the United States general population with a standard deviation of 10 because calibration testing was performed on a large sample of the general population. You can read more about the calibration and centering samples on HealthMeasures.net (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>). The T-score is provided with an error term (Standard Error or SE). The Standard Error is a statistical measure of variance and represents the "margin of error" for the T-score.

Important: *A higher PROMIS T-score represents more of the concept being measured.* For negatively-worded concepts like depression, a T-score of 60 is one SD worse than average. By comparison, a depression T-score of 40 is one SD better than average.

Standard Error (SE): A PROMIS score includes a T-score and a standard error (SE). The standard error is a measure of the variability for a given T-score across hypothetical repeated measurements. The standard error can be used to construct confidence intervals around a T-score. A 95% confidence interval is common. A 95%

confidence interval means there is a 95% probability that the true T-score is within this range. The formula for a 95% confidence interval is $(T\text{-score} \pm (1.96 * SE))$. For example, if $T=52$ and $SE=2$, the lower boundary of the confidence interval is $(52 - (1.96 * 2)) = 48$ and the upper boundary is $(52 + (1.96 * 2)) = 56$.

FREQUENTLY ASKED QUESTIONS (FAQs)

Q: I am interested in learning more. Where can I do that?

Review the HealthMeasures website at www.healthmeasures.net.

Q: How do I handle multiple responses when administering a short form on paper?

Guidelines on how to deal with multiple responses have been established. Resolution depends on the responses noted by the research participant.

- If two or more responses are marked by the respondent, and they are next to one another, then a data entry specialist will be responsible for randomly selecting one of them to be entered and will write down on the form which answer was selected. Note: To randomly select one of two responses, the data entry specialist will flip a coin (heads - higher number will be entered; tails – lower number will be entered). To randomly select one of three (or more) responses, a table of random numbers should be used with a statistician's assistance.
- If two or more responses are marked, and they are NOT all next to one another, the response will be considered missing.

Q: What is the minimum change on a PROMIS instrument that represents a clinically meaningful difference?

To learn more about research on the meaning of a change in scores, we suggest conducting a literature review to identify the most current information. The HealthMeasures website (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>) has additional information on interpreting scores.

APPENDIX 1 — SCORING TABLES

PROMIS Adult Short Form v1.0 – Depression 4a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
4	41.0	6.2
5	49.0	3.2
6	51.8	2.7
7	53.9	2.4
8	55.7	2.3
9	57.3	2.3
10	58.9	2.3
11	60.5	2.3
12	62.2	2.3
13	63.9	2.3
14	65.7	2.3
15	67.5	2.3
16	69.4	2.3
17	71.2	2.4
18	73.3	2.4
19	75.7	2.6
20	79.4	2.6

SE* = Standard Error on T-Score

PROMIS Adult Short Form v1.0 – Depression 6a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
6	38.4	5.8
7	45.2	3.4
8	48.3	2.8
9	50.4	2.4
10	52.0	2.2
11	53.4	2.1
12	54.7	2.0
13	55.9	2.0
14	57.0	1.9
15	58.2	1.9
16	59.3	2.0
17	60.5	2.0
18	61.7	2.0
19	62.9	2.0
20	64.2	2.0
21	65.5	2.0
22	66.7	2.0
23	68.0	2.0
24	69.3	2.0
25	70.6	2.0
26	72.0	2.0
27	73.4	2.0
28	75.0	2.1
29	76.9	2.4
30	80.3	3.5

SE* = Standard Error on T-Score

PROMIS Adult Short Form v1.0 – Depression 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
8	38.2	5.7
9	44.7	3.3
10	47.5	2.7
11	49.4	2.3
12	50.9	2.0
13	52.1	1.9
14	53.2	1.8
15	54.1	1.8
16	55.1	1.7
17	55.9	1.7
18	56.8	1.7
19	57.7	1.7
20	58.5	1.7
21	59.4	1.7
22	60.3	1.7
23	61.2	1.7
24	62.1	1.8
25	63.0	1.8
26	63.9	1.8
27	64.9	1.8
28	65.8	1.8
29	66.8	1.8
30	67.7	1.8
31	68.7	1.8
32	69.7	1.8
33	70.7	1.8
34	71.7	1.8
35	72.8	1.8
36	73.9	1.8
37	75.0	1.9
38	76.4	2.0
39	78.2	2.4
40	81.3	3.4

SE* = Standard Error on T-Score

PROMIS Adult Short Form v1.0 – Depression 8b		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
8	37.1	5.5
9	43.3	3.4
10	46.2	2.8
11	48.2	2.4
12	49.8	2.2
13	51.2	2.0
14	52.3	1.9
15	53.4	1.8
16	54.3	1.8
17	55.3	1.7
18	56.2	1.7
19	57.1	1.7
20	57.9	1.7
21	58.8	1.7
22	59.7	1.8
23	60.7	1.8
24	61.6	1.8
25	62.5	1.8
26	63.5	1.8
27	64.4	1.8
28	65.4	1.8
29	66.4	1.8
30	67.4	1.8
31	68.3	1.8
32	69.3	1.8
33	70.4	1.8
34	71.4	1.8
35	72.5	1.8
36	73.6	1.8
37	74.8	1.9
38	76.2	2.0
39	77.9	2.4
40	81.1	3.4

SE* = Standard Error on T-Score

PROMIS Adult Short Form v1.0 – Depression-OA-Knee 4a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
4	37.8	5.7
5	44.4	3.7
6	47.5	3.4
7	49.9	3.1
8	51.9	3.0
9	53.9	3.0
10	55.7	3.0
11	57.6	3.0
12	59.5	3.0
13	61.4	3.0
14	63.4	3.1
15	65.5	3.0
16	67.5	3.0
17	69.6	3.1
18	72.1	3.1
19	74.9	3.3
20	78.6	3.9

*SE = Standard Error on T-score metric

PROMIS Pediatric Short Form GenPop v3.0 – Depressive Symptoms 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
8	39.9	5.9
9	46.9	3.2
10	49.3	2.7
11	51.0	2.3
12	52.4	2.1
13	53.6	2.0
14	54.6	1.9
15	55.6	1.8
16	56.5	1.8
17	57.4	1.8
18	58.3	1.8
19	59.1	1.8
20	60.0	1.8
21	60.8	1.8
22	61.7	1.8
23	62.5	1.8
24	63.3	1.8
25	64.1	1.8
26	64.9	1.8
27	65.7	1.8
28	66.5	1.7
29	67.3	1.7
30	68.0	1.7
31	68.8	1.7
32	69.6	1.7
33	70.4	1.7
34	71.2	1.8
35	72.1	1.8
36	73.1	1.9
37	74.2	2.0
38	75.5	2.2
39	77.2	2.5
40	80.3	3.4

*SE = Standard Error on T-score metric

PROMIS Early Childhood Parent-Report Short Form v1.0 – Depressive Symptoms 4a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
4	42.4	6.8
5	50.5	4.4
6	54.5	3.8
7	57.6	3.5
8	60.2	3.4
9	62.5	3.4
10	64.8	3.4
11	67.0	3.4
12	69.1	3.4
13	70.9	3.4
14	72.7	3.4
15	74.6	3.3
16	76.4	3.3
17	78.4	3.3
18	80.4	3.3
19	82.4	3.2
20	84.9	3.1

*SE = T-score standard error

PROMIS Early Childhood Parent-Report Short Form v1.0 – Depressive Symptoms 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
8	41.0	6.5
9	47.8	4.4
10	51.0	3.7
11	53.5	3.1
12	55.4	2.8
13	57.1	2.6
14	58.5	2.5
15	59.9	2.5
16	61.2	2.4
17	62.4	2.4
18	63.6	2.4
19	64.8	2.4
20	65.9	2.4
21	67.1	2.4
22	68.2	2.4
23	69.3	2.4
24	70.3	2.4
25	71.3	2.4
26	72.3	2.4
27	73.3	2.4
28	74.3	2.3
29	75.3	2.3
30	76.2	2.3
31	77.2	2.3
32	78.2	2.3
33	79.2	2.3
34	80.2	2.3
35	81.2	2.3
36	82.3	2.4
37	83.5	2.4
38	84.8	2.4
39	86.1	2.3
40	87.4	2.0

SE* = Standard Error on T-Score

PROMIS Parent Proxy Short Form GenPop v3.0 – Depressive Symptoms 6a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
6	40.8	6.2
7	48.2	3.5
8	51.1	2.9
9	53.2	2.5
10	54.9	2.4
11	56.4	2.3
12	57.9	2.2
13	59.2	2.2
14	60.6	2.2
15	61.9	2.2
16	63.2	2.2
17	64.6	2.2
18	65.9	2.2
19	67.1	2.2
20	68.3	2.2
21	69.6	2.2
22	70.7	2.1
23	71.9	2.1
24	73.0	2.1
25	74.2	2.1
26	75.4	2.2
27	76.7	2.3
28	78.2	2.5
29	79.8	2.6
30	82.7	3.1

*SE = Standard Error on T-score metric

APPENDIX 2 – SCORING TABLES FOR RETIRED MEASURES

PROMIS Pediatric Short Form v1.0 – Depressive Symptoms 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
0	35.2	5.8
1	40.4	4.6
2	43.2	4.2
3	45.5	3.8
4	47.4	3.6
5	49.1	3.4
6	50.5	3.3
7	51.9	3.2
8	53.2	3.1
9	54.3	3.1
10	55.5	3.0
11	56.6	3.0
12	57.7	3.0
13	58.7	3.0
14	59.7	2.9
15	60.7	2.9
16	61.8	2.9
17	62.8	2.9
18	63.7	2.9
19	64.7	2.9
20	65.7	2.9
21	66.8	2.9
22	67.8	2.9
23	68.8	2.9
24	69.9	2.9
25	70.9	2.9
26	72.1	3.0
27	73.2	3.0
28	74.5	3.1
29	75.9	3.2
30	77.5	3.4
31	79.3	3.5
32	81.9	3.7

SE* = Standard Error on T-Score

PROMIS Pediatric Short Form v1.1 – Depressive Symptoms 8b		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
0	35.2	5.8
1	40.4	4.6
2	43.2	4.2
3	45.5	3.9
4	47.4	3.7
5	49.1	3.5
6	50.6	3.3
7	52.0	3.2
8	53.3	3.2
9	54.5	3.1
10	55.7	3.1
11	56.8	3.0
12	57.9	3.0
13	59.0	3.0
14	60.0	3.0
15	61.1	3.0
16	62.1	3.0
17	63.1	3.0
18	64.1	3.0
19	65.1	3.0
20	66.1	3.0
21	67.2	2.9
22	66.2	2.9
23	69.3	3.0
24	70.3	3.0
25	71.4	3.0
26	72.6	3.0
27	73.8	3.1
28	75.1	3.2
29	76.5	3.3
30	78.1	3.5
31	79.9	3.6
32	82.4	3.7

SE*= Standard Error on T-Score

PROMIS Pediatric Short Form v2.0 – Depressive Symptoms 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
8	35.2	5.8
9	40.4	4.6
10	43.2	4.2
11	45.5	3.9
12	47.4	3.7
13	49.1	3.5
14	50.6	3.3
15	52.0	3.2
16	53.3	3.2
17	54.5	3.1
18	55.7	3.1
19	56.8	3.0
20	57.9	3.0
21	59.0	3.0
22	60.0	3.0
23	61.1	3.0
24	62.1	3.0
25	63.1	3.0
26	64.1	3.0
27	65.1	3.0
28	66.1	3.0
29	67.2	2.9
30	68.2	2.9
31	69.3	3.0
32	70.3	3.0
33	71.4	3.0
34	72.6	3.0
35	73.8	3.1
36	75.1	3.2
37	76.5	3.3
38	78.1	3.5
39	79.9	3.6
40	82.4	3.7

SE* = Standard Error on T-Score

PROMIS Parent Proxy Short Form v1.0 – Depressive Symptoms 6a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
0	36.0	6.0
1	42.0	4.0
2	45.0	4.0
3	48.0	4.0
4	50.0	3.0
5	52.0	3.0
6	54.0	3.0
7	55.0	3.0
8	57.0	3.0
9	59.0	3.0
10	60.0	3.0
11	62.0	3.0
12	64.0	3.0
13	65.0	3.0
14	67.0	3.0
15	68.0	3.0
16	70.0	3.0
17	72.0	3.0
18	73.0	3.0
19	75.0	3.0
20	77.0	3.0
21	78.0	3.0
22	80.0	3.0
23	83.0	4.0
24	86.0	4.0

SE* = Standard Error on T-Score

PROMIS Parent Proxy Short Form v1.1 – Depressive Symptoms 6b		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
0	36.0	5.9
1	42.1	4.4
2	45.4	4.0
3	48.0	3.6
4	50.2	3.4
5	52.1	3.3
6	53.9	3.2
7	55.6	3.1
8	57.2	3.1
9	58.9	3.1
10	60.5	3.1
11	62.1	3.1
12	63.7	3.1
13	65.3	3.1
14	66.8	3.1
15	68.4	3.1
16	70.0	3.1
17	71.5	3.0
18	73.1	3.0
19	74.7	3.0
20	76.3	3.1
21	78.1	3.2
22	80.2	3.3
23	82.5	3.4
24	84.7	3.2

SE* = Standard Error on T-Score

PROMIS Parent Proxy** Short Form v2.0 – Depressive Symptoms		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
6	36.2	5.9
7	42.1	4.4
8	45.4	4.0
9	48.0	3.6
10	50.2	3.4
11	52.1	3.3
12	53.9	3.2
13	55.6	3.1
14	57.2	3.1
15	58.9	3.1
16	60.5	3.1
17	62.1	3.1
18	63.7	3.1
19	65.3	3.1
20	66.8	3.1
21	68.4	3.1
22	70.0	3.1
23	71.5	3.0
24	73.1	3.0
25	74.7	3.0
26	76.3	3.1
27	78.1	3.2
28	80.2	3.3
29	82.5	3.4
30	84.7	3.2

SE* = Standard Error on T-Score

**All scoring tables are based on default Parent Proxy calibrations.

APPENDIX 3 – CROSSWALK BETWEEN PEDIATRIC AND PARENT PROXY V2.0 T-SCORES AND PEDIATRIC AND PARENT PROXY GENPOP V3.0 T-SCORES

Table 1. PROMIS Pediatric Depressive Symptoms: Crosswalk between v2.0 T-scores and GenPop v3.0 T-scores

PROMIS Pediatric Depressive Symptoms T-Score		PROMIS Pediatric Depressive Symptoms T-Score	
v2.0	v3.0 GenPop	v2.0	v3.0 GenPop
32	37.0	70	69.5
33	41.0	71	70.0
34	41.7	72	70.3
35	42.2	73	71.5
36	43.7	74	72.3
37	44.6	75	73.0
38	45.6	76	74.0
39	46.5	77	74.5
40	47.3	78	75.0
41	48.2	79	76.0
42	49.0	80	76.0
43	49.5	81	77.0
44	50.0		
45	50.7		
46	51.5		
47	52.1		
48	52.7		
49	53.5		
50	54.2		
51	54.9		
52	55.6		
53	56.2		
54	56.8		
55	57.5		
56	58.2		
57	59.0		
58	60.0		
59	60.9		
60	61.6		
61	62.2		
62	63.4		
63	64.1		
64	64.6		
65	65.4		
66	66.0		
67	66.5		
68	68.0		
69	68.7		

Table 2. PROMIS Parent Proxy Depressive Symptoms: Crosswalk between v2.0 T-scores and GenPop v3.0 T-scores

PROMIS Parent Proxy Depressive Symptoms T-Score		PROMIS Parent Proxy Depressive Symptoms T-Score	
v2.0	v3.0 GenPop	v2.0	v3.0 GenPop
32	37.0	70	70.0
33	39.5	71	71.6
34	40.0	72	71.9
35	41.2	73	72.3
36	43.3	74	73.0
37	43.7	75	74.0
38	44.4	76	74.0
39	45.4	77	75.0
40	46.4		
41	47.7		
42	48.3		
43	49.3		
44	50.0		
45	50.6		
46	51.2		
47	52.1		
48	52.9		
49	53.6		
50	54.3		
51	55.1		
52	55.9		
53	56.6		
54	57.3		
55	58.1		
56	58.9		
57	59.6		
58	60.5		
59	61.2		
60	62.1		
61	63.0		
62	63.7		
63	64.3		
64	64.8		
65	66.0		
66	67.0		
67	67.4		
68	68.6		
69	69.3		

ANXIETY

SCORING MANUAL

A brief guide to scoring the PROMIS® Anxiety instruments:

ADULT	PEDIATRIC	EARLY CHILDHOOD PARENT-REPORT	PARENT PROXY
PROMIS Bank v1.0 – Anxiety PROMIS Bank v1.0 – Anxiety (recommended) PROMIS Bank v1.0 – Anxiety (screen-to-CAT) PROMIS Short Form v1.0 – Anxiety 4a PROMIS Short Form v1.0 – Anxiety 6a PROMIS Short Form v1.0 – Anxiety 7a PROMIS Short Form v1.0 – Anxiety 8a PROMIS-Ca Bank v1.0 – Anxiety	PROMIS Pediatric Bank GenPop v3.0 – Anxiety PROMIS Pediatric Bank GenPop v3.0 – Anxiety (screen-to-CAT) PROMIS Pediatric Short Form GenPop v3.0 – Anxiety 8a PROMIS Pediatric Bank v2.0 – Anxiety* PROMIS Pediatric Bank v2.0 – Anxiety (recommended)* PROMIS Pediatric Bank v2.0 – Anxiety (screen-to-CAT)* PROMIS Pediatric Short Form v2.0 – Anxiety 8a* PROMIS Pediatric Bank v1.1 – Anxiety* PROMIS Pediatric Short Form v1.1 – Anxiety 8b* PROMIS Pediatric Bank v1.0 – Anxiety* PROMIS Pediatric Short Form v1.0 – Anxiety 8a*	PROMIS Early Childhood Parent-Report Bank v1.0 – Anxiety PROMIS Early Childhood Parent-Report Short Form v1.0 – Anxiety 4a PROMIS Early Childhood Parent-Report Short Form v1.0 – Anxiety 8a	PROMIS Parent Proxy Bank GenPop v3.0 – Anxiety PROMIS Parent Proxy Bank GenPop v3.0 – Anxiety (screen-to-CAT) PROMIS Parent Proxy Short Form GenPop v3.0 – Anxiety 8a PROMIS Parent Proxy Bank v2.0 – Anxiety* PROMIS Parent Proxy Bank v2.0 – Anxiety (recommended)* PROMIS Parent Proxy Bank v2.0 – Anxiety (screen-to-CAT)* PROMIS Parent Proxy Short Form v2.0 – Anxiety 8a* PROMIS Parent Proxy Bank v1.1 – Anxiety* PROMIS Parent Proxy Short Form v1.1 – Anxiety 8b* PROMIS Parent Proxy Bank v1.0 – Anxiety* PROMIS Parent Proxy Short Form v1.0 – Anxiety 8a*

*Retired measure

COMPARING SCORES ACROSS VERSIONS

Some PROMIS domains have multiple versions of instruments (i.e., v1.0, v1.1, v2.0, v3.0). Generally, **it is recommended that you use the most recent version available which can be identified as the instrument with the highest version number**. In most cases, an instrument that has a decimal increase (v1.0 to v1.1) retains the same item-level parameters as well as instrument reliability and validity. In cases where a version number increases by a whole number (e.g., v1.0 to v2.0), the changes to the instrument are more substantial.

- For Anxiety Pediatric and Parent Proxy measures, the calibrations between v1.0, v1.1, and v2.0 are identical. Consequently, T-scores from v1.0, v1.1, and v2.0 are comparable.
- Scores from Pediatric and Parent Proxy GenPop v3.0 are NOT comparable with v1.0, v1.1, or v2.0 measure scores.

Transforming Scores between Versions:

To transform T-scores from Pediatric/Parent Proxy v1.0/v1.1/v2.0 Anxiety measures to the Pediatric/Parent Proxy GenPop v3.0 metric, rescore v1.0/v1.1/v2.0 measures using the [HealthMeasures Scoring Service](#) or Assessment Center API. This is the preferred strategy as it produces precise scores.

- Download the v3.0 measure from [Search & View Measures](#) on HealthMeasures.net.
- Revise your dataset as needed to match the v3.0 item IDs and response scores:

- **PROMIS Pediatric Bank GenPop v3.0 – Anxiety, PROMIS Pediatric Bank GenPop v3.0 – Anxiety (screen-to-CAT), and PROMIS Pediatric Short Form GenPop v3.0 – Anxiety 8a:**
 - The Item ID **2230R1r** was changed to **2230R1r2** to account for the revised Item Stem from "I got scared really easy." to "I got scared really easily."
- **PROMIS Parent Proxy Bank GenPop v3.0 – Anxiety, PROMIS Parent Proxy Bank GenPop v3.0 – Anxiety (screen-to-CAT), and PROMIS Parent Proxy Short Form GenPop v3.0 – Anxiety 8a:**
 - The Item ID **Pf1anxiety1r** was changed to **Pf1anxiety1r2** to account for the revised Item Stem from "My child got scared really easy." to "My child got scared really easily."
- **PROMIS Parent Proxy Bank GenPop v3.0 – Anxiety and PROMIS Parent Proxy Bank GenPop v3.0 – Anxiety (screen-to-CAT):**
 - The Item ID **Pf2anxiety3r** was changed to **Pf2anxiety3r2** to account for the revised Response Scores from 1-2-3-4-4 to 1-2-3-4-5.

3. Follow the instructions for using the [HealthMeasures Scoring Service](https://www.healthmeasures.net/scoring-service). Contact help@HealthMeasures.net if using the Assessment Center API for rescoring.

The crosswalk tables in Appendix 3 may also be used as a reference to guide understanding of how scores have shifted from the older v1.0, v1.1, and v2.0 measures to the v3.0 measures. If these tables are used to convert older T-scores to GenPop v3.0, please note that the resulting T-scores are not as precise as those generated by using HealthMeasures Scoring Service or Assessment Center API.

WHICH CALIBRATION SAMPLE SHOULD I USE?

Some PROMIS Parent Proxy v1.0, v1.1, and v2.0 measures (Anxiety, Depressive Symptoms, Fatigue, Mobility, Pain Interference, Peer Relationships) had two calibration samples – “Parent Proxy” and “Parent Proxy Without Local Dependence.” The former (Parent Proxy) included calibrations for all items. This was the default calibration sample. The Parent Proxy Without Local Dependence did not include calibrations for some items. The items without calibrations are enemy items. That is, a dyad or triad of items was identified in which there are psychometric reasons to only administer one of those items to a given respondent. For example, item Pf2anxiety5 and Pf2anxiety4 are enemy items. A participant should only see one of these items in a CAT. The v3.0 GenPop measure have a single calibration sample and no enemy items.

SCORING THE INSTRUMENT

Short Forms: PROMIS instruments are scored using item-level calibrations. This means that the most accurate way to score a PROMIS instrument is to use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoring-service) or a data collection tool that automatically calculates scores (e.g., REDCap auto-score). This method of scoring uses responses to each item for each participant. We refer to this as “response pattern scoring.” Response pattern scoring is preferred because it is more accurate than the use of raw score/scale score look up tables included in this manual. Response pattern scoring is especially useful when there is missing data (i.e., a respondent skipped an item), different groups of participants responded to different items, or you have created a new questionnaire using a subset of questions from a PROMIS item bank.

Instructions for the HealthMeasures Scoring Service

1. Download the Input Template from the [HealthMeasures Scoring Service](https://www.healthmeasures.net/scoring-service) home page.

2. Add your data (all respondents' individual answers to questions) for one PROMIS measure to the template (multiple timepoints are okay).
3. Determine how much data is missing. In order to produce a precise score, a respondent must answer at least 4 items in measures that include 4 or more items. If a precise score is required, remove respondents who answered less than 4 items from the input template and do not calculate a T-score.
4. Upload your spreadsheet to the HealthMeasures Scoring Service. If this is your first time, register as a new user.
5. Access your email to receive a spreadsheet with calculated scores.

Instructions for Using Raw Sum Score to T-score Conversion Tables

Each question usually has five response options ranging in value from one to five. To find the total raw score for a short form with all questions answered, sum the values of the response to each question. For example, for the adult 8-item form, the lowest possible raw score is 8; the highest possible raw score is 40 (see all short form scoring tables in Appendix 1). **All questions must be answered in order to produce a valid score using the scoring tables.** If a participant has skipped a question, use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoringervice) to generate a final score.

Locate the applicable score conversion table in Appendix 1 and use this table to translate the total raw score into a T-score for each participant. The T-score rescales the raw score into a standardized T-score with a mean of 50 and a standard deviation (SD) of 10. Therefore, a person with a T-score of 40 is one SD below the mean.

For the adult PROMIS Anxiety 7a short form, a raw score of 10 converts to a T-score of 46.7 with a standard error (SE) of 2.6 (see scoring table for the 7a short form in Appendix 1). Thus, the 95% confidence interval around the observed score ranges from 40.6 to 51.7 (T-score $\pm (1.96*SE)$ or $46.7 \pm (1.96*2.6)$).

CAT: A minimum number of items (e.g., 4) must be answered in order to receive a score for the Anxiety CATs. The response to the first item will guide the system's choice of the next item for the participant. The participant's response to the second item will dictate the selection of the following question, and so on. As additional items are administered, the potential for error is reduced and confidence in the respondent's score increases. The CAT will continue until either the standard error drops below a specified level (e.g., on the T-score metric 3.0), or the participant has answered the maximum number of questions (e.g., 12), whichever occurs first. For some CATs, specifically "recommended" and "screen-to-CAT" there are additional stopping rules. These include stopping when the standard error isn't improving much or if a respondent is asymptomatic. For details on the exact stopping rules for Anxiety CATs, view the Measure Differences summary.

SCORES

For most PROMIS instruments, a score of 50 is the average for the United States general population with a standard deviation of 10 because calibration testing was performed on a large sample of the general population. You can read more about the calibration and centering samples on HealthMeasures.net (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>). The T-score is provided with an error term (Standard Error or SE). The Standard Error is a statistical measure of variance and represents the "margin of error" for the T-score.

Important: *A higher PROMIS T-score represents more of the concept being measured.* For negatively-worded concepts like anxiety, a T-score of 60 is one SD worse than average. By comparison, an anxiety T-score of 40 is one SD better than average.

Standard Error (SE): A PROMIS score includes a T-score and a standard error (SE). The standard error is a measure of the variability for a given T-score across hypothetical repeated measurements. The standard error can be used to construct confidence intervals around a T-score. A 95% confidence interval is common. A 95% confidence interval means there is a 95% probability that the true T-score is within this range. The formula for a 95% confidence interval is $(T\text{-score} \pm (1.96 * SE))$. For example, if $T=52$ and $SE=2$, the lower boundary of the confidence interval is $52 - (1.96 * 2) = 48$ and the upper boundary is $52 + (1.96 * 2) = 56$.

FREQUENTLY ASKED QUESTIONS (FAQs)

Q: I am interested in learning more. Where can I do that?

Review the HealthMeasures website at www.healthmeasures.net.

Q: How do I handle multiple responses when administering a short form on paper?

Guidelines on how to deal with multiple responses have been established. Resolution depends on the responses noted by the research participant.

- If two or more responses are marked by the respondent, and they are next to one another, then a data entry specialist will be responsible for randomly selecting one of them to be entered and will write down on the form which answer was selected. Note: To randomly select one of two responses, the data entry specialist will flip a coin (heads - higher number will be entered; tails – lower number will be entered). To randomly select one of three (or more) responses, a table of random numbers should be used with a statistician's assistance.
- If two or more responses are marked, and they are NOT all next to one another, the response will be considered missing.

Q: What is the minimum change on a PROMIS instrument that represents a clinically meaningful difference?

To learn more about research on the meaning of a change in scores, we suggest conducting a literature review to identify the most current information. The HealthMeasures website (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>) has additional information on interpreting scores.

APPENDIX 1 – SCORING TABLES

PROMIS Adult Short Form v1.0 – Anxiety 4a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
4	40.3	6.1
5	48	3.6
6	51.2	3.1
7	53.7	2.8
8	55.8	2.7
9	57.7	2.6
10	59.5	2.6
11	61.4	2.6
12	63.4	2.6
13	65.3	2.7
14	67.3	2.7
15	69.3	2.7
16	71.2	2.7
17	73.3	2.7
18	75.4	2.7
19	77.9	2.9
20	81.6	3.7

SE* = Standard Error on T-Score

PROMIS Adult Short Form v1.0 – Anxiety 6a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
6	39.1	5.9
7	45.9	3.4
8	48.8	2.9
9	50.9	2.6
10	52.7	2.4
11	54.2	2.3
12	55.6	2.2
13	56.9	2.2
14	58.2	2.2
15	59.4	2.2
16	60.7	2.2
17	62.0	2.2
18	63.3	2.2
19	64.6	2.2
20	66.0	2.2
21	67.3	2.2
22	68.6	2.2
23	70.0	2.2
24	71.3	2.2
25	72.7	2.2
26	74.1	2.2
27	75.6	2.3
28	77.4	2.4
29	79.4	2.7
30	82.7	3.5

SE* = Standard Error on T-Score

PROMIS Adult Short Form v1.0 – Anxiety 7a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
7	36.3	5.4
8	42.1	3.4
9	44.7	2.9
10	46.7	2.6
11	48.4	2.4
12	49.9	2.3
13	51.3	2.3
14	52.6	2.2
15	53.8	2.2
16	55.1	2.2
17	56.3	2.2
18	57.6	2.2
19	58.8	2.2
20	60.0	2.2
21	61.3	2.2
22	62.6	2.2
23	63.8	2.2
24	65.1	2.2
25	66.4	2.2
26	67.7	2.2
27	68.9	2.2
28	70.2	2.2
29	71.5	2.2
30	72.9	2.2
31	74.3	2.2
32	75.8	2.3
33	77.4	2.4
34	79.5	2.7
35	82.7	3.5

SE* = Standard Error on T-Score

PROMIS Adult Short Form v1.0 – Anxiety 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
8	37.1	5.5
9	43.2	3.3
10	45.9	2.8
11	47.8	2.5
12	49.4	2.3
13	50.8	2.2
14	52.1	2.1
15	53.2	2.0
16	54.3	2.0
17	55.4	2.0
18	56.4	2.0
19	57.4	2.0
20	58.4	2.0
21	59.4	2.0
22	60.4	2.0
23	61.4	2.0
24	62.5	2.0
25	63.5	2.0
26	64.5	2.0
27	65.6	2.0
28	66.6	2.0
29	67.7	2.0
30	68.7	2.0
31	69.8	2.0
32	70.8	2.0
33	71.9	2.0
34	73.0	2.0
35	74.1	2.0
36	75.4	2.0
37	76.7	2.1
38	78.2	2.3
39	80.0	2.6
40	83.1	3.4

SE* = Standard Error on T-Score

PROMIS Pediatric Short Form GenPop v3.0 – Anxiety 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
8	39.0	5.8
9	45.4	3.3
10	47.8	2.8
11	49.6	2.4
12	51.0	2.2
13	52.2	2.1
14	53.3	2.0
15	54.4	2.0
16	55.3	1.9
17	56.3	1.9
18	57.2	1.9
19	58.1	1.9
20	59.0	1.9
21	59.9	1.9
22	60.8	1.9
23	61.7	2.0
24	62.6	2.0
25	63.4	1.9
26	64.3	1.9
27	65.1	1.9
28	65.9	1.9
29	66.8	1.9
30	67.6	1.9
31	68.4	1.9
32	69.2	1.9
33	70.0	1.9
34	70.9	1.9
35	71.8	2.0
36	72.8	2.1
37	73.9	2.2
38	75.2	2.4
39	76.7	2.6
40	79.8	3.4

*SE = Standard Error on T-score metric

PROMIS Parent Proxy Short Form GenPop v3.0 – Anxiety 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
8	38.8	5.8
9	45.2	3.5
10	48.0	2.8
11	49.9	2.5
12	51.5	2.2
13	52.8	2.1
14	54.0	2.1
15	55.2	2.0
16	56.3	2.0
17	57.3	2.0
18	58.4	2.0
19	59.4	2.0
20	60.4	2.0
21	61.4	2.0
22	62.5	2.0
23	63.4	2.0
24	64.4	2.0
25	65.3	2.0
26	66.3	2.0
27	67.2	2.0
28	68.1	2.0
29	69.0	2.0
30	69.9	2.0
31	70.8	1.9
32	71.7	1.9
33	72.6	2.0
34	73.5	2.0
35	74.5	2.0
36	75.6	2.1
37	76.8	2.2
38	78.2	2.4
39	80.0	2.6
40	82.7	3.0

*SE = Standard Error on T-score metric

PROMIS Early Childhood Parent-Report Short Form v1.0 – Anxiety 4a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
4	40.9	6.5
5	47.8	4.9
6	51.6	4.6
7	54.9	4.1
8	57.5	4.0
9	60.0	3.9
10	62.5	4.0
11	65.0	3.9
12	67.4	3.9
13	69.6	4.0
14	71.9	4.0
15	74.3	3.9
16	76.6	3.8
17	78.8	3.7
18	81.1	3.7
19	83.2	3.5
20	85.2	3.1

SE* = Standard Error on T-Score

PROMIS Early Childhood Parent-Report Short Form v1.0 – Anxiety 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
8	39.6	6.2
9	45.6	4.4
10	48.6	3.9
11	51.0	3.3
12	52.9	3.0
13	54.5	2.8
14	56.0	2.7
15	57.4	2.7
16	58.8	2.7
17	60.1	2.7
18	61.4	2.7
19	62.7	2.7
20	64.0	2.7
21	65.3	2.7
22	66.5	2.7
23	67.8	2.7
24	69.0	2.7
25	70.3	2.7
26	71.6	2.7
27	72.8	2.7
28	74.1	2.6
29	75.3	2.6
30	76.4	2.5
31	77.6	2.5
32	78.7	2.5
33	79.9	2.5
34	81.1	2.6
35	82.3	2.6
36	83.6	2.6
37	84.9	2.6
38	86.0	2.4
39	87.0	2.2
40	87.9	1.9

SE* = Standard Error on T-Score

APPENDIX 2 – SCORING TABLES FOR RETIRED MEASURES

PROMIS Pediatric Short Form v1.0 – Anxiety 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
0	32.3	5.7
1	36.7	4.9
2	39.2	4.7
3	41.4	4.3
4	43.3	4.2
5	45.1	4.0
6	46.7	3.9
7	48.2	3.8
8	49.6	3.8
9	50.9	3.7
10	52.3	3.7
11	53.5	3.7
12	54.8	3.7
13	56.0	3.7
14	57.3	3.7
15	58.5	3.7
16	59.7	3.7
17	60.9	3.7
18	62.1	3.7
19	63.3	3.7
20	64.5	3.7
21	65.8	3.7
22	67.0	3.7
23	68.3	3.7
24	69.6	3.7
25	70.9	3.7
26	72.3	3.7
27	73.7	3.7
28	75.2	3.8
29	76.8	3.9
30	78.6	4.0
31	80.5	4.0
32	82.8	3.9

*SE = Standard Error on T-score

PROMIS Pediatric Short Form v1.1 – Anxiety 8b		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
0	33.5	5.9
1	38.0	4.9
2	40.6	4.7
3	43.0	4.4
4	44.9	4.2
5	46.7	4.0
6	48.3	3.9
7	49.8	3.8
8	51.2	3.8
9	52.5	3.7
10	53.8	3.7
11	55.1	3.7
12	56.3	3.7
13	57.5	3.7
14	58.7	3.7
15	59.9	3.7
16	61.0	3.7
17	62.2	3.7
18	63.4	3.7
19	64.5	3.7
20	65.7	3.6
21	66.9	3.6
22	68.1	3.6
23	69.3	3.7
24	70.6	3.7
25	71.8	3.7
26	73.2	3.7
27	74.6	3.8
28	76.0	3.8
29	77.6	3.0
30	79.3	4.0
31	81.1	3.9
32	83.3	3.8

*SE = Standard Error on T-score

PROMIS Parent Proxy Short Form v1.0 – Anxiety 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-score	SE*
0	34.0	6.0
1	38.0	5.0
2	41.0	4.0
3	44.0	4.0
4	46.0	4.0
5	48.0	3.0
6	49.0	3.0
7	51.0	3.0
8	52.0	3.0
9	54.0	3.0
10	55.0	3.0
11	56.0	3.0
12	58.0	3.0
13	59.0	3.0
14	61.0	3.0
15	62.0	3.0
16	64.0	3.0
17	65.0	3.0
18	66.0	3.0
19	68.0	3.0
20	69.0	3.0
21	71.0	3.0
22	72.0	3.0
23	73.0	3.0
24	75.0	3.0
25	76.0	3.0
26	77.0	3.0
27	79.0	3.0
28	80.0	3.0
29	82.0	3.0
30	84.0	3.0
31	86.0	4.0
32	88.0	4.0

*SE = Standard Error on T-score

PROMIS Parent Proxy Short Form v1.1 – Anxiety 8b		
Short Form Conversion Table		
Raw Summed Score	T-score	SE*
0	34.6	5.6
1	39.7	4.4
2	42.7	3.9
3	45.1	3.5
4	47.1	3.3
5	48.8	3.1
6	50.4	3.0
7	51.9	2.9
8	53.3	2.9
9	54.7	2.9
10	56.0	2.9
11	57.4	2.9
12	58.7	2.9
13	60.1	2.9
14	61.4	2.9
15	62.8	2.9
16	64.2	3.0
17	65.6	3.0
18	66.9	3.0
19	68.3	3.0
20	69.3	2.9
21	71.0	2.9
22	72.3	2.9
23	73.6	2.8
24	74.9	2.8
25	76.1	2.8
26	77.4	2.8
27	78.8	2.8
28	80.2	2.9
29	81.6	2.9
30	83.2	2.9
31	84.8	2.8
32	86.4	2.6

*SE = Standard Error on T-score

PROMIS Pediatric Short Form v2.0 – Anxiety 8a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
8	33.5	5.9
9	38.0	4.9
10	40.6	4.7
11	43.0	4.4
12	44.9	4.2
13	46.7	4.0
14	48.3	3.9
15	49.8	3.8
16	51.2	3.8
17	52.5	3.7
18	53.8	3.7
19	55.1	3.7
20	56.3	3.7
21	57.5	3.7
22	58.7	3.7
23	59.9	3.7
24	61.0	3.7
25	62.2	3.7
26	63.4	3.7
27	64.5	3.7
28	65.7	3.6
29	66.9	3.6
30	68.1	3.6
31	69.3	3.7
32	70.6	3.7
33	71.8	3.7
34	73.2	3.7
35	74.6	3.8
36	76.0	3.8
37	77.6	3.9
38	79.3	4.0
39	81.1	3.9
40	83.3	3.8

SE* = Standard Error on T-Score

PROMIS Parent Proxy** Short Form v2.0 – Anxiety 8a					
<i>Short Form Conversion Table</i>					
Raw Summed Score	T-Score	SE*	Raw Summed Score	T-Score	SE*
8	34.6	5.6	25	65.6	3.0
9	39.7	4.4	26	66.9	3.0
10	42.7	3.9	27	68.3	3.0
11	45.1	3.5	28	69.6	2.9
12	47.1	3.3	29	71.0	2.9
13	48.8	3.1	30	72.3	2.9
14	50.4	3.0	31	73.6	2.8
15	51.9	2.9	32	74.9	2.8
16	53.3	2.9	33	76.1	2.8
17	54.7	2.9	34	77.4	2.8
18	56.0	2.8	35	78.8	2.8
19	57.4	2.9	36	80.2	2.9
20	58.7	2.9	37	81.6	2.9
21	60.1	2.9	38	83.2	2.9
22	61.4	2.9	39	84.8	2.8
23	62.8	2.9	40	86.4	2.6
24	64.2	3.0			

SE* = Standard Error on T-Score

**All scoring tables are based on default Parent Proxy calibrations.

APPENDIX 3 – CROSSWALK BETWEEN PEDIATRIC AND PARENT PROXY V2.0 T-SCORES AND PEDIATRIC AND PARENT PROXY GENPOP V3.0 T-SCORES

Table 1. PROMIS Pediatric Anxiety: Crosswalk between v2.0 T-scores and GenPop v3.0 T-scores

PROMIS Pediatric Anxiety T-Score		PROMIS Pediatric Anxiety T-Score	
v2.0	v3.0 GenPop	v2.0	v3.0 GenPop
32	37.0	70	68.5
33	41.5	71	69.4
34	41.5	72	70.2
35	42.1	73	70.8
36	43.4	74	71.0
37	44.2	75	71.7
38	44.8	76	74.5
39	46.0	77	74.5
40	46.7	78	74.5
41	47.5	79	74.5
42	48.2	80	74.5
43	48.9	81	74.5
44	49.5	82	77.0
45	50.1		
46	50.8		
47	51.5		
48	52.1		
49	52.7		
50	53.5		
51	54.1		
52	54.9		
53	55.6		
54	56.0		
55	56.8		
56	57.8		
57	58.5		
58	59.1		
59	60.1		
60	61.0		
61	61.8		
62	62.4		
63	63.3		
64	64.1		
65	64.8		
66	65.4		
67	66.3		
68	67.0		
69	67.9		

Table 2. PROMIS Parent Proxy Anxiety: Crosswalk between v2.0 T-scores and GenPop v3.0 T-scores

PROMIS Parent Proxy Anxiety T-Score		PROMIS Parent Proxy Anxiety T-Score	
v2.0	v3.0 GenPop	v2.0	v3.0 GenPop
34	38.0	72	69.7
35	42.5	73	70.0
36	42.5	74	70.8
37	42.8	75	71.5
38	43.8	76	73.0
39	44.5	77	73.0
40	45.5	78	73.0
41	46.3	79	74.7
42	46.9		
43	47.8		
44	48.6		
45	49.4		
46	50.1		
47	50.9		
48	51.5		
49	52.3		
50	53.2		
51	53.9		
52	54.7		
53	55.4		
54	56.1		
55	56.9		
56	57.8		
57	58.5		
58	59.2		
59	60.0		
60	60.8		
61	61.8		
62	62.6		
63	63.4		
64	64.0		
65	64.7		
66	65.3		
67	65.9		
68	66.3		
69	66.4		
70	67.0		
71	68.4		

SOCIAL ISOLATION SCORING MANUAL

A brief guide to scoring the PROMIS® Social Isolation instruments:

ADULT
PROMIS Item Bank v2.0 – Social Isolation
PROMIS Short Form v2.0 – Social Isolation 4a
PROMIS Short Form v2.0 – Social Isolation 6a
PROMIS Short Form v2.0 – Social Isolation 8a
PROMIS Short Form v2.0 – Social Isolation-OA-Knee 2a

OA=osteoarthritis

SCORING THE INSTRUMENT

Short Forms: PROMIS instruments are scored using item-level calibrations. This means that the most accurate way to score a PROMIS instrument is to use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoringervice) or a data collection tool that automatically calculates scores (e.g., REDCap auto-score). This method of scoring uses responses to each item for each participant. We refer to this as “response pattern scoring.” Because response pattern scoring is more accurate than the use of raw score/scale score look up tables included in this manual, it is preferred. Response pattern scoring is especially useful when there is missing data (i.e., a respondent skipped an item), different groups of participants responded to different items, or you have created a new questionnaire using a subset of questions from a PROMIS item bank.

To use the scoring tables in this manual, calculate a summed score. Each question has five response options ranging in value from one to five. To find the total raw score for a short form with all questions answered, sum the values of the response to each question. For example, for the 4a short form, the lowest possible raw summed score is 4; the highest possible raw summed score is 20. **All questions must be answered in order to produce a valid score using the scoring tables.** If a participant has skipped a question, use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoringervice) to generate a final score.

With the total raw score for a measure, locate the applicable score conversion table in the Appendix and use this table to translate the total raw score into a T-score for each participant. The T-score rescales the raw score into a standardized score with a mean of 50 and a standard deviation (SD) of 10. Therefore, a person with a T-score of 40 is one SD below the mean.

For the Social Isolation 8a short form, a raw score of 10 converts to a T-score of 41.4 with a standard error (SE) of 2.4 (see scoring table for the 8a short form in Appendix). Thus, the 95% confidence interval around the observed score ranges from 37.0 to 45.8 ($T\text{-score} \pm (1.96 * SE)$ or $41.4 \pm (1.96 * 2.4)$).

CAT: A minimum number of items must be answered in order to receive a score for Social Isolation CAT. The first item is selected because it provides the most information about the U.S. general population. The response to this item will guide the system’s choice of the next item for the participant. The participant’s response to this item will dictate the selection of the following question, and so on. As additional items are administered, the potential for error is reduced and confidence in the respondent’s score increases. CAT will continue until either the standard error drops below a specified level, or the participant has answered the maximum number of questions (12), whichever occurs first.

SCORES

For most PROMIS instruments, a score of 50 is the average for the United States general population with a standard deviation of 10 because calibration testing was performed on a large sample of the general population. You can read more about the calibration and centering samples on HealthMeasures.net (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>). The T-score is provided with an error term (Standard Error or SE). The Standard Error is a statistical measure of variance and represents the “margin of error” for the T-score.

Important: *A higher PROMIS T-score represents more of the concept being measured.* For negatively-worded concepts like Social Isolation, a T-score of 60 is one SD worse than average. By comparison, a Social Isolation T-score of 40 is one SD better than average.

Standard Error (SE): A PROMIS score includes a T-score and a standard error (SE). The standard error is a measure of the variability for a given T-score across hypothetical repeated measurements. The standard error can be used to construct confidence intervals around a T-score. A 95% confidence interval is common. A 95% confidence interval means there is a 95% probability that the true T-score is within this range. The formula for a 95% confidence interval is $(T\text{-score} \pm (1.96 * SE))$. For example, if $T=52$ and $SE=2$, the lower boundary of the confidence interval is $(52 - (1.96 * 2)) = 48$ and the upper boundary is $(52 + (1.96 * 2)) = 56$.

FREQUENTLY ASKED QUESTIONS (FAQ)

Q: I am interested in learning more. Where can I do that?

Review the HealthMeasures website at www.healthmeasures.net.

Q: How do I handle multiple responses when administering a short form on paper?

Guidelines on how to deal with multiple responses have been established. Resolution depends on the responses noted by the research participant.

- If two or more responses are marked by the respondent, and they are next to one another, then a data entry specialist will be responsible for randomly selecting one of them to be entered and will write down on the form which answer was selected. Note: To randomly select one of two responses, the data entry specialist will flip a coin (heads - higher number will be entered; tails – lower number will be entered). To randomly select one of three (or more) responses, a table of random numbers should be used with a statistician’s assistance.
- If two or more responses are marked, and they are NOT all next to one another, the response will be considered missing.

Q: What is the minimum change on a PROMIS instrument that represents a clinically meaningful difference?

To learn more about research on the meaning of a change in scores, we suggest conducting a literature review to identify the most current information. The HealthMeasures website (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>) has additional information on interpreting scores.

APPENDIX—SCORING TABLES

In the tables below, “Raw Score” = sum of response scores. “Scale score” is a T-score.

Social Isolation 4a Short Form Conversion Table		
Raw Score	Scale Score	SE*
4	34.8	5.1
5	40.4	3.2
6	43.3	2.8
7	45.7	2.7
8	47.8	2.6
9	49.8	2.6
10	51.8	2.6
11	53.9	2.6
12	56.1	2.6
13	58.1	2.7
14	60.1	2.6
15	62.0	2.6
16	63.8	2.5
17	65.5	2.6
18	67.5	2.7
19	69.9	2.9
20	74.2	4.2

*SE = Standard Error on T-score metric

Adult version

Social Isolation 6a Short Form Conversion Table		
Raw Score	Scale Score	SE*
6	34.4	5.0
7	39.7	3.1
8	42.2	2.6
9	44.2	2.3
10	45.8	2.2
11	47.3	2.2
12	48.7	2.2
13	50.1	2.2
14	51.5	2.2
15	53.0	2.2
16	54.4	2.2
17	55.9	2.2
18	57.3	2.3
19	58.8	2.3
20	60.2	2.2
21	61.5	2.2
22	62.8	2.2
23	64.0	2.2
24	65.2	2.1
25	66.5	2.2
26	67.7	2.2
27	69.1	2.3
28	70.8	2.6
29	72.6	2.8
30	76.2	4.0

*SE = Standard Error on T-score metric

Social Isolation 8a Short Form Conversion Table		
Raw Score	Scale Score	SE*
8	33.9	4.9
9	39.1	3.0
10	41.4	2.4
11	43.1	2.1
12	44.4	2.0
13	45.7	1.9
14	46.8	1.9
15	47.9	1.9
16	48.9	1.9
17	50.0	1.8
18	51.0	1.8
19	52.0	1.9
20	53.1	1.9
21	54.2	1.9
22	55.3	1.9
23	56.4	1.9
24	57.5	1.9
25	58.6	1.9
26	59.6	1.9
27	60.7	1.9
28	61.7	1.9
29	62.6	1.8
30	63.6	1.8
31	64.5	1.8
32	65.4	1.8
33	66.3	1.8
34	67.2	1.8
35	68.2	1.9
36	69.2	2.0
37	70.4	2.1
38	71.8	2.4
39	73.4	2.6
40	76.9	3.9

*SE = Standard Error on T-score metric

Adult version

PROMIS Adult Short Form v2.0 – Social Isolation-OA-Knee 2a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
2	37.3	5.6
3	44.5	3.6
4	48.5	3.3
5	52.2	3.4
6	56.1	3.4
7	59.8	3.4
8	63.1	3.4
9	66.5	3.5
10	72.0	4.5

*SE = Standard Error on T-score metric



EMOTIONAL SUPPORT SCORING MANUAL

A brief guide to scoring the PROMIS[®] Emotional Support instruments:

ADULT
PROMIS Item Bank v2.0 – Emotional Support
PROMIS Short Form v2.0 – Emotional Support 4a
PROMIS Short Form v2.0 – Emotional Support 6a
PROMIS Short Form v2.0 – Emotional Support 8a

SCORING THE INSTRUMENT

Short Forms: PROMIS instruments are scored using item-level calibrations. This means that the most accurate way to score a PROMIS instrument is to use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoring-service) or a data collection tool that automatically calculates scores (e.g., REDCap auto-score). This method of scoring uses responses to each item for each participant. We refer to this as “response pattern scoring.” Because response pattern scoring is more accurate than the use of raw score/scale score look up tables included in this manual, it is preferred. Response pattern scoring is especially useful when there is missing data (i.e., a respondent skipped an item), different groups of participants responded to different items, or you have created a new questionnaire using a subset of questions from a PROMIS item bank.

To use the scoring tables in this manual, calculate a summed score. Each question usually has five response options ranging in value from one to five. To find the total raw score for a short form with all questions answered, sum the values of the response to each question. For example, for the v2.0 adult 4-item form, the lowest possible raw score is 4; the highest possible raw score is 16 (see all short form scoring tables in Appendix). **All questions must be answered in order to produce a valid score using the scoring tables.** If a participant has skipped a question, use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoring-service) to generate a final score.

With the total raw score for a measure, locate the applicable score conversion table in the Appendix and use this table to translate the total raw score into a T-score for each participant. The T-score rescales the raw score into a standardized score with a mean of 50 and a standard deviation (SD) of 10. Therefore a person with a T-score of 40 is one SD below the mean.

For the adult PROMIS Emotional Support 4a short form v2.0, a raw score of 10 converts to a T-score of 38.9 with a standard error (SE) of 2.1 (see scoring table for the 4a v2.0 short form in the Appendix). Thus, the 95% confidence interval around the observed score ranges from 34.8 to 43.0 (T-score $\pm$ (1.96*SE) or 38.9 $\pm$ (1.96*2.1)).

CAT: A minimum number of items (4 for adult CATs) must be answered in order to receive a score for the Emotional Support CAT. The response to the first item will guide the system’s choice of the next item for the participant. The participant’s response to the second item will dictate the selection of the following question, and so on. As additional items are administered, the potential for error is reduced and confidence in the respondent’s score increases. CAT will continue until either the standard error drops below a specified level (on the T-score metric 3.0 for adult and adult cancer CATs and 4.0 for peds and parent proxy CATs), or the participant has answered the maximum number of questions (12), whichever occurs first.



SCORES

For most PROMIS instruments, a score of 50 is the average for the United States general population with a standard deviation of 10 because calibration testing was performed on a large sample of the general population. You can read more about the calibration and centering samples on HealthMeasures.net (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>). The T-score is provided with an error term (Standard Error or SE). The Standard Error is a statistical measure of variance and represents the “margin of error” for the T-score.

Important: *A higher PROMIS T-score represents more of the concept being measured.* For positively-worded concepts like Emotional Support, a T-score of 60 is one SD better than average. By comparison, an Emotional Support T-score of 40 is one SD worse than average.

Standard Error (SE): A PROMIS score includes a T-score and a standard error (SE). The standard error is a measure of the variability for a given T-score across hypothetical repeated measurements. The standard error can be used to construct confidence intervals around a T-score. A 95% confidence interval is common. A 95% confidence interval means there is a 95% probability that the true T-score is within this range. The formula for a 95% confidence interval is $(T\text{-score} \pm (1.96 * SE))$. For example, if $T=52$ and $SE=2$, the lower boundary of the confidence interval is $(52 - (1.96 * 2)) = 48$ and the upper boundary is $(52 + (1.96 * 2)) = 56$.

FREQUENTLY ASKED QUESTIONS (FAQs)

Q: I am interested in learning more. Where can I do that?

Review the HealthMeasures website at www.healthmeasures.net.

Q: How do I handle multiple responses when administering a short form on paper?

Guidelines on how to deal with multiple responses have been established. Resolution depends on the responses noted by the research participant.

- If two or more responses are marked by the respondent, and they are next to one another, then a data entry specialist will be responsible for randomly selecting one of them to be entered and will write down on the form which answer was selected. Note: To randomly select one of two responses, the data entry specialist will flip a coin (heads - higher number will be entered; tails – lower number will be entered). To randomly select one of three (or more) responses, a table of random numbers should be used with a statistician’s assistance.
- If two or more responses are marked, and they are NOT all next to one another, the response will be considered missing.

Q: What is the minimum change on a PROMIS instrument that represents a clinically meaningful difference?

To learn more about research on the meaning of a change in scores, we suggest conducting a literature review to identify the most current information. The HealthMeasures website (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>) has additional information on interpreting scores.

APPENDIX-SCORING TABLES

In the tables below, “Raw Score” = sum of response scores. “Scale score” is a T-score.

Emotional Support 4a Short Form Conversion Table		
Raw Score	Scale Score	SE*
4	25.7	4.0
5	29.9	2.6
6	32.1	2.3
7	34.0	2.2
8	35.7	2.1
9	37.3	2.1
10	38.9	2.1
11	40.5	2.1
12	42.1	2.1
13	43.7	2.1
14	45.4	2.2
15	47.2	2.2
16	49.0	2.2
17	50.8	2.2
18	53.0	2.3
19	55.6	2.8
20	62.0	5.5

*SE = Standard Error on T-score metric

Adult version

Emotional Support 6a Short Form Conversion Table		
Raw Score	Scale Score	SE*
6	25.0	3.9
7	28.9	2.4
8	30.7	2.0
9	32.2	1.9
10	33.5	1.8
11	34.6	1.7
12	35.7	1.7
13	36.8	1.7
14	37.8	1.7
15	38.9	1.7
16	40.0	1.7
17	41.1	1.7
18	42.2	1.7
19	43.3	1.7
20	44.4	1.7
21	45.5	1.8
22	46.7	1.8
23	48.0	1.8
24	49.2	1.8
25	50.5	1.8
26	51.8	1.8
27	53.3	1.9
28	54.9	2.1
29	57.2	2.7
30	63.2	5.2

*SE = Standard Error on T-score metric

Emotional Support 8a Short Form Conversion Table		
Raw Score	Scale Score	SE*
8	24.7	3.8
9	28.4	2.3
10	30.1	1.9
11	31.4	1.7
12	32.5	1.6
13	33.5	1.5
14	34.3	1.5
15	35.2	1.5
16	36.0	1.5
17	36.8	1.5
18	37.6	1.5
19	38.4	1.5
20	39.2	1.5
21	39.9	1.5
22	40.7	1.5
23	41.5	1.5
24	42.3	1.5
25	43.1	1.5
26	43.9	1.5
27	44.7	1.5
28	45.6	1.5
29	46.4	1.5
30	47.3	1.6
31	48.2	1.6
32	49.2	1.6
33	50.1	1.6
34	51.1	1.6
35	52.1	1.6
36	53.1	1.6
37	54.3	1.7
38	55.7	2.0
39	57.8	2.6
40	63.5	5.2

*SE = Standard Error on T-score metric

Adult version

COMPANIONSHIP SCORING MANUAL

A brief guide to scoring the PROMIS® Companionship instruments:

ADULT
PROMIS Short Form v2.0 – Companionship 4a
PROMIS Short Form v2.0 – Companionship 6a

SCORING THE INSTRUMENT

Short Forms: PROMIS instruments are scored using item-level calibrations. This means that the most accurate way to score a PROMIS instrument is to use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoring-service) or a data collection tool that automatically calculates scores (e.g., REDCap auto-score). This method of scoring uses responses to each item for each participant. We refer to this as “response pattern scoring.” Because response pattern scoring is more accurate than the use of raw score/scale score look up tables included in this manual, it is preferred. Response pattern scoring is especially useful when there is missing data (i.e., a respondent skipped an item), different groups of participants responded to different items, or you have created a new questionnaire using a subset of questions from a PROMIS item bank.

To use the scoring tables in this manual, calculate a summed score. Each question usually has five response options ranging in value from one to five. To find the total raw score for a short form with all questions answered, sum the values of the response to each question. For example, for the v2.0 adult 4-item form, the lowest possible raw score is 4; the highest possible raw score is 16 (see all short form scoring tables in Appendix 1). **All questions must be answered in order to produce a valid score using the scoring tables.** If a participant has skipped a question, use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoring-service) to generate a final score.

With the total raw score for a measure, locate the applicable score conversion table in the Appendix and use this table to translate the total raw score into a T-score for each participant. The T-score rescales the raw score into a standardized score with a mean of 50 and a standard deviation (SD) of 10. Therefore a person with a T-score of 40 is one SD below the mean.

For the adult PROMIS Companionship 4a short form, a raw score of 10 converts to a T-score of 39.1 with a standard error (SE) of 2.2 (see scoring table for the 4a short form in Appendix). Thus, the 95% confidence interval around the observed score ranges from 34.8 to 43.4 (T-score $\pm (1.96*SE)$ or $39.1 \pm (1.96*2.2)$).

SCORES

For most PROMIS instruments, a score of 50 is the average for the United States general population with a standard deviation of 10 because calibration testing was performed on a large sample of the general population. You can read more about the calibration and centering samples on HealthMeasures.net (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>). The T-score is provided with an error term (Standard Error or SE). The Standard Error is a statistical measure of variance and represents the “margin of error” for the T-score.

Important *A higher PROMIS T-score represents more of the concept being measured.* For positively-worded concepts like Companionship, a T-score of 60 is one SD better than average. By comparison, a Companionship T-score of 40 is one SD worse than average.

Standard Error (SE): A PROMIS score includes a T-score and a standard error (SE). The standard error is a measure of the variability for a given T-score across hypothetical repeated measurements. The standard error can be used to construct confidence intervals around a T-score. A 95% confidence interval is common. A 95% confidence interval means there is a 95% probability that the true T-score is within this range. The formula for a 95% confidence interval is $(T\text{-score} \pm (1.96 * SE))$. For example, if $T=52$ and $SE=2$, the lower boundary of the confidence interval is $(52 - (1.96 * 2)) = 48$ and the upper boundary is $(52 + (1.96 * 2)) = 56$.

FREQUENTLY ASKED QUESTIONS (FAQs)

Q: I am interested in learning more. Where can I do that?

Review the HealthMeasures website at www.healthmeasures.net.

Q: How do I handle multiple responses when administering a short form on paper?

Guidelines on how to deal with multiple responses have been established. Resolution depends on the responses noted by the research participant.

- If two or more responses are marked by the respondent, and they are next to one another, then a data entry specialist will be responsible for randomly selecting one of them to be entered and will write down on the form which answer was selected. Note: To randomly select one of two responses, the data entry specialist will flip a coin (heads - higher number will be entered; tails – lower number will be entered). To randomly select one of three (or more) responses, a table of random numbers should be used with a statistician's assistance.
- If two or more responses are marked, and they are NOT all next to one another, the response will be considered missing.

Q: What is the minimum change on a PROMIS instrument that represents a clinically meaningful difference?

To learn more about research on the meaning of a change in scores, we suggest conducting a literature review to identify the most current information. The HealthMeasures website (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>) has additional information on interpreting scores.

APPENDIX-SCORING TABLES

In the tables below, “Raw Score” = sum of response scores. “Scale score” is a T-score.

Companionship 4a <i>Short Form Conversion Table</i>		
Raw Score	Scale Score	SE*
4	25.2	3.9
5	29.5	2.5
6	31.8	2.2
7	33.8	2.2
8	35.5	2.1
9	37.3	2.2
10	39.1	2.2
11	40.8	2.2
12	42.5	2.2
13	44.3	2.2
14	46.2	2.2
15	48.1	2.2
16	50.0	2.2
17	52.1	2.3
18	54.3	2.3
19	56.9	2.7
20	63.1	5.3
*SE = Standard Error on T-score metric		

Companionship 6a <i>Short Form Conversion Table</i>		
Raw Score	Scale Score	SE*
6	24.2	3.9
7	28.0	2.6
8	30.1	2.3
9	31.7	2.2
10	33.1	2.1
11	34.4	2.1
12	35.7	2.1
13	36.9	2.1
14	38.1	2.2
15	39.3	2.2
16	40.5	2.2
17	41.6	2.2
18	42.7	2.2
19	43.9	2.2
20	45.1	2.3
21	46.3	2.3
22	47.6	2.3
23	48.8	2.3
24	50.1	2.3
25	51.4	2.3
26	52.9	2.3
27	54.6	2.4
28	56.4	2.6
29	58.7	3.1
30	64.2	5.2
*SE = Standard Error on T-score metric		

ABILITY TO PARTICIPATE IN SOCIAL ROLES AND ACTIVITIES

SCORING MANUAL

A brief guide to scoring the PROMIS[®] Ability to Participate in Social Roles and Activities v2.0 instruments:

ADULT
PROMIS Item Bank v2.0 – Ability to Participate in Social Roles and Activities
PROMIS Item Bank v2.0 – Ability to Participate in Social Roles and Activities (recommended)
PROMIS Item Bank v2.0 – Ability to Participate in Social Roles and Activities (screen-to-CAT)
PROMIS Short Form v2.0 – Ability to Participate in Social Roles and Activities 4a
PROMIS Short Form v2.0 – Ability to Participate in Social Roles and Activities 6a
PROMIS Short Form v2.0 – Ability to Participate in Social Roles and Activities 8a
PROMIS Short Form v2.0 – Ability to Participate in Social Roles and Activities-OA-Knee 8a

OA=osteoarthritis

SCORING THE INSTRUMENT

Short Forms: PROMIS instruments are scored using item-level calibrations. This means that the most accurate way to score a PROMIS instrument is to use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoring-service) or a data collection tool that automatically calculates scores (e.g., REDCap auto-score). This method of scoring uses responses to each item for each participant. We refer to this as “response pattern scoring.” Because response pattern scoring is more accurate than the use of raw score/scale score look up tables included in this manual, it is preferred. Response pattern scoring is especially useful when there is missing data (i.e., a respondent skipped an item), different groups of participants responded to different items, or you have created a new questionnaire using a subset of questions from a PROMIS item bank.

To use the scoring tables in this manual, calculate a summed score. Each question usually has five response options ranging in value from one to five. To find the total raw score for a short form with all questions answered, sum the values of the response to each question. For example, for the v2.0 adult 4-item form, the lowest possible raw score is 4; the highest possible raw score is 20 (see all short form scoring tables in the Appendix). **All questions must be answered in order to produce a valid score using the scoring tables.** If a participant has skipped a question, use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoring-service) to generate a final score.

With the total raw score for a measure, locate the applicable score conversion table in the Appendix and use this table to translate the total raw score into a T-score for each participant. The T-score rescales the raw score into a standardized score with a mean of 50 and a standard deviation (SD) of 10. Therefore, a person with a T-score of 40 is one SD below the mean.

For the adult PROMIS Ability to Participate in Social Roles and Activities 8a short form, a raw score of 10 converts to a T-score of 31.3 with a standard error (SE) of 1.9 (see scoring table for the 8a short form in appendix). Thus, the 95% confidence interval around the observed score ranges from 27.6 to 35.0 (T-score $\pm$ (1.96*SE) or 31.3 $\pm$ (1.96*1.9)).

CAT: A minimum number of items (e.g., 4) must be answered in order to receive a score for the Ability to Participate in Social Roles and Activities CAT. The response to the first item will guide the system’s choice of the next item for the participant. The participant’s response to the second item will dictate the selection of the

following question, and so on. As additional items are administered, the potential for error is reduced and confidence in the respondent's score increases. The CAT will continue until either the standard error drops below a specified level (e.g., on the T-score metric 3.0), or the participant has answered the maximum number of questions (e.g., 12), whichever occurs first. For some CATs, specifically "recommended" and "screen-to-CAT" there are additional stopping rules. These include stopping when the standard error isn't improving much or if a respondent is asymptomatic. For details on the exact stopping rules for Ability to Participate in Social Roles and Activities CATs, view the Measure Differences summary.

SCORES

For most PROMIS instruments, a score of 50 is the average for the United States general population with a standard deviation of 10 because calibration testing was performed on a large sample of the general population. You can read more about the calibration and centering samples on HealthMeasures.net (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>). The T-score is provided with an error term (Standard Error or SE). The Standard Error is a statistical measure of variance and represents the "margin of error" for the T-score.

Important: *A higher PROMIS T-score represents more of the concept being measured.* For positively-worded concepts like Ability to Participate in Social Roles and Activities, a T-score of 60 is one SD better than average. By comparison, an Ability to Participate in Social Roles and Activities T-score of 40 is one SD worse than average.

Standard Error (SE): A PROMIS score includes a T-score and a standard error (SE). The standard error is a measure of the variability for a given T-score across hypothetical repeated measurements. The standard error can be used to construct confidence intervals around a T-score. A 95% confidence interval is common. A 95% confidence interval means there is a 95% probability that the true T-score is within this range. The formula for a 95% confidence interval is $(T\text{-score} \pm (1.96 * SE))$. For example, if $T=52$ and $SE=2$, the lower boundary of the confidence interval is $(52 - (1.96 * 2)) = 48$ and the upper boundary is $(52 + (1.96 * 2)) = 56$.

FREQUENTLY ASKED QUESTIONS (FAQs)

Q: I am interested in learning more. Where can I do that?

Review the HealthMeasures website at www.healthmeasures.net.

Q: How do I handle multiple responses when administering a short form on paper?

Guidelines on how to deal with multiple responses have been established. Resolution depends on the responses noted by the research participant.

- If two or more responses are marked by the respondent, and they are next to one another, then a data entry specialist will be responsible for randomly selecting one of them to be entered and will write down on the form which answer was selected. Note: To randomly select one of two responses, the data entry specialist will flip a coin (heads - higher number will be entered; tails – lower number will be entered). To randomly select one of three (or more) responses, a table of random numbers should be used with a statistician's assistance.
- If two or more responses are marked, and they are NOT all next to one another, the response will be considered missing.



Q: What is the minimum change on a PROMIS instrument that represents a clinically meaningful difference?
To learn more about research on the meaning of a change in scores, we suggest conducting a literature review to identify the most current information. The HealthMeasures website (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>) has additional information on interpreting scores.

APPENDIX—SCORING TABLES

Ability to Participate Social 4a Short Form Conversion Table		
Raw Score	Scale Score	SE*
4	27.5	4.1
5	31.8	2.5
6	34.0	2.3
7	35.7	2.2
8	37.3	2.1
9	38.8	2.2
10	40.5	2.3
11	42.3	2.3
12	44.2	2.3
13	46.2	2.3
14	48.1	2.2
15	50.0	2.2
16	51.9	2.2
17	53.7	2.3
18	55.8	2.3
19	58.3	2.7
20	64.2	5.1
*SE = Standard Error on T-score metric		

Adult version

Ability to Participate Social 6a Short Form Conversion Table		
Raw Score	Scale Score	SE*
6	26.7	4.0
7	30.7	2.4
8	32.5	2.0
9	33.9	1.9
10	35.1	1.8
11	36.2	1.8
12	37.2	1.8
13	38.2	1.8
14	39.3	1.8
15	40.4	1.8
16	41.6	1.9
17	42.9	1.9
18	44.2	1.9
19	45.6	1.9
20	46.9	1.9
21	48.2	1.8
22	49.5	1.8
23	50.7	1.8
24	51.9	1.8
25	53.2	1.8
26	54.5	1.9
27	55.9	1.9
28	57.5	2.1
29	59.6	2.6
30	65.0	4.9
*SE = Standard Error on T-score metric		

Ability to Participate Social 8a Short Form Conversion Table		
Raw Score	Scale Score	SE*
8	25.9	3.9
9	29.7	2.3
10	31.3	1.9
11	32.6	1.7
12	33.6	1.6
13	34.5	1.6
14	35.3	1.5
15	36.2	1.5
16	36.9	1.5
17	37.7	1.5
18	38.5	1.5
19	39.3	1.6
20	40.2	1.6
21	41.1	1.6
22	42.0	1.7
23	43.0	1.7
24	44.0	1.7
25	45.0	1.7
26	46.0	1.6
27	47.0	1.6
28	48.0	1.6
29	48.9	1.6
30	49.9	1.6
31	50.8	1.6
32	51.7	1.6
33	52.7	1.6
34	53.6	1.6
35	54.6	1.6
36	55.7	1.6
37	56.8	1.7
38	58.2	2.0
39	60.2	2.5
40	65.4	4.9
*SE = Standard Error on T-score metric		

Adult version

PROMIS Adult Short Form v2.0 – Ability to Participate in Social Roles and Activities-OA-Knee 8a					
Short Form Conversion Table					
Raw Summed Score	T-Score	SE*	Raw Summed Score	T-Score	SE*
8	26.1	3.9	25	45.5	1.8
9	29.9	2.5	26	46.5	1.8
10	31.6	2.1	27	47.5	1.7
11	32.9	1.9	28	48.4	1.7
12	34.0	1.8	29	49.4	1.7
13	35.0	1.7	30	50.3	1.7
14	35.9	1.7	31	51.2	1.7
15	36.7	1.7	32	52.2	1.7
16	37.5	1.7	33	53.1	1.7
17	38.4	1.7	34	54.1	1.7
18	39.2	1.7	35	55.1	1.7
19	40.0	1.7	36	56.2	1.8
20	40.9	1.7	37	57.4	1.9
21	41.8	1.7	38	58.8	2.2
22	42.7	1.8	39	60.9	2.7
23	43.6	1.8	40	66.0	4.8
24	44.6	1.8			

*SE = Standard Error on T-score metric

SPIRITUALITY SCORING MANUAL

A brief guide to scoring the PROMIS[®] Spirituality instruments:

ADULT
PROMIS Bank v1.0 – Spirituality PROMIS Short Form v1.0– Spirituality 6a PROMIS Short Form v1.0 – Spirituality – Personal Strength 6b PROMIS Short Form v1.0 – Spirituality – Religiosity 6c

COMPARING SCORES ACROSS VERSIONS

Some PROMIS domains have multiple versions of instruments (i.e., v1.0, v1.1, v2.0, v3.0). Generally, **it is recommended that you use the most recent version available which can be identified as the instrument with the highest version number**. In most cases, an instrument that has a decimal increase (v1.0 to v1.1) retains the same item-level parameters as well as instrument reliability and validity. In cases where a version number increases by a whole number (e.g., v1.0 to v2.0), the changes to the instrument are more substantial.

Adult

There is only one version (v1.0) of the Spirituality instruments.

SCORING THE INSTRUMENT

Short Forms: PROMIS instruments are scored using item-level calibrations. This means that the most accurate way to score a PROMIS instrument is to use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoringervice) or a data collection tool that automatically calculates scores (e.g., REDCap auto-score). This method of scoring uses responses to each item for each participant. We refer to this as “response pattern scoring.” Because response pattern scoring is more accurate than the use of raw score/scale score look up tables included in this manual, it is preferred. Response pattern scoring is especially useful when there is missing data (i.e., a respondent skipped an item), different groups of participants responded to different items, or you have created a new short form using a subset of questions from a PROMIS item bank.

To use the scoring tables in this manual, calculate a summed score. Each question has five response options ranging in value from one to five. To find the total raw summed score for a short form with all questions answered, sum the values of the response to each question. For example, for the v1.0 Spirituality 6a short form, the lowest possible raw score is 6; the highest possible raw score is 30 (see all short form scoring tables in Appendix 1). **All questions must be answered in order to produce a valid score using the scoring tables.** If a respondent has skipped a question, use the HealthMeasures Scoring Service (https://www.assessmentcenter.net/ac_scoringervice) to generate a final score.

With the total raw summed score for a measure, locate the applicable score conversion table in Appendix 1 and use this table to translate the total raw score into a T-score for each participant. The T-score rescales the raw score into a standardized score with a mean of 50 and a standard deviation (SD) of 10. Therefore, a person with a T-score of 40 is one SD below the mean. For example, for the adult PROMIS Spirituality 6a short form v1.0, a

raw score of 10 converts to a T-score of 40.5 with a standard error (SE) of 2.1 (see scoring table for the 6a v1.0 short form in Appendix 1).

CAT: A minimum number of items (i.e., 4) must be answered in order to receive a score for the Spirituality CAT. The response to the first item will guide the system's choice of the next item for the participant. The participant's response to the second item will dictate the selection of the following question, and so on. As additional items are administered, the potential for error is reduced and confidence in the respondent's score increases. The CAT will continue until either the standard error drops below a specified level (on the T-score metric 3.0), or the participant has answered the maximum number of questions (8), whichever occurs first. For details on the exact stopping rules for Spirituality CAT, view the Measure Differences summary.

SCORES

For most PROMIS instruments, a score of 50 is the average for the United States general population with a standard deviation of 10 because calibration testing was performed on a large sample of the general population. However, the Spirituality measures were developed in a clinical sample of individuals who reported receiving care from a healthcare provider. A mean of 50 reflects the average for that sample. You can read more about the calibration and centering samples on HealthMeasures.net (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>). The T-score is provided with an error term (Standard Error or SE). The Standard Error is a statistical measure of variance and represents the "margin of error" for the T-score.

Important: *A higher PROMIS T-score represents more of the concept being measured.* For positively-worded concepts like Spirituality, a T-score of 60 is one SD more than average. By comparison, a Spirituality T-score of 40 is one SD less than average.

Standard Error (SE): A PROMIS score includes a T-score and a standard error (SE). The standard error is a measure of the variability for a given T-score across hypothetical repeated measurements. The standard error can be used to construct confidence intervals around a T-score. A 95% confidence interval is common. A 95% confidence interval means there is a 95% probability that the true T-score is within this range. The formula for a 95% confidence interval is $(T\text{-score} \pm (1.96 * SE))$. For example, if $T=52$ and $SE=2$, the lower boundary of the confidence interval is $(52 - (1.96 * 2)) = 48$ and the upper boundary is $(52 + (1.96 * 2)) = 56$.

FREQUENTLY ASKED QUESTIONS (FAQ)

Q: I am interested in learning more. Where can I do that?

Review the HealthMeasures website at www.healthmeasures.net. You can also find information specifically about the HEAL measures at <https://www.HealingFactors.pitt.edu>.

Q: How do I handle multiple responses when administering a short form on paper?

Guidelines on how to deal with multiple responses have been established. Resolution depends on the responses noted by the research participant.

- If two or more responses are marked by the respondent, and they are next to one another, then a data entry specialist will be responsible for randomly selecting one of them to be entered and will write down on the form which answer was selected. Note: To randomly select one of two responses, the data entry specialist will flip a coin (heads - higher number will be entered; tails – lower number will be entered).



To randomly select one of three (or more) responses, a table of random numbers should be used with a statistician's assistance.

- If two or more responses are marked, and they are NOT all next to one another, the response will be considered missing.

Q: What is the minimum change on a PROMIS instrument that represents a clinically meaningful difference?

To learn more about research on the meaning of a change in scores, we suggest conducting a literature review to identify the most current information. The HealthMeasures website (<http://www.healthmeasures.net/score-and-interpret/interpret-scores/promis>) has additional information on interpreting scores.

APPENDIX 1 — SCORING TABLES

PROMIS Adult SF v1.0 – Spirituality 6a		
<i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
6	29.2	5.7
7	35.4	2.9
8	37.6	2.4
9	39.2	2.2
10	40.5	2.1
11	41.7	2.0
12	42.8	2.0
13	43.8	2.0
14	44.8	1.9
15	45.8	1.9
16	46.7	1.9
17	47.7	1.9
18	48.6	1.9
19	49.5	1.9
20	50.4	1.9
21	51.3	1.9
22	52.3	1.9
23	53.2	2.0
24	54.2	2.0
25	55.3	2.0
26	56.4	2.1
27	57.7	2.2
28	59.3	2.5
29	61.5	3.0
30	68.6	6.3
*SE = Standard Error on T-score metric		

PROMIS Adult SF v1.0 – Spirituality – Personal Strength 6b <i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
6	28.1	5.9
7	34.3	3.1
8	36.6	2.7
9	38.4	2.4
10	39.8	2.2
11	41.1	2.2
12	42.3	2.1
13	43.4	2.1
14	44.5	2.1
15	45.5	2.1
16	46.6	2.1
17	47.6	2.1
18	48.6	2.1
19	49.6	2.1
20	50.6	2.1
21	51.7	2.1
22	52.7	2.1
23	53.7	2.1
24	54.8	2.1
25	56.0	2.2
26	57.3	2.3
27	58.8	2.6
28	60.6	3.0
29	63.1	3.5
30	70.1	6.5
*SE = Standard Error on T-score metric		

PROMIS Adult SF v1.0 – Spirituality – Religiosity 6c <i>Short Form Conversion Table</i>		
Raw Summed Score	T-Score	SE*
6	30.7	7.6
7	38.4	4.7
8	41.0	4.3
9	43.1	3.9
10	44.8	3.8
11	46.4	3.4
12	47.7	3.3
13	49.0	3.2
14	50.2	3.1
15	51.3	3.1
16	52.4	3.1
17	53.5	3.1
18	54.6	3.1
19	55.7	3.1
20	56.8	3.1
21	58.0	3.2
22	59.2	3.2
23	60.4	3.3
24	61.7	3.4
25	63.2	3.5
26	64.8	3.8
27	66.6	4.0
28	68.9	4.4
29	71.1	4.6
30	77.4	6.8
*SE = Standard Error on T-score metric		